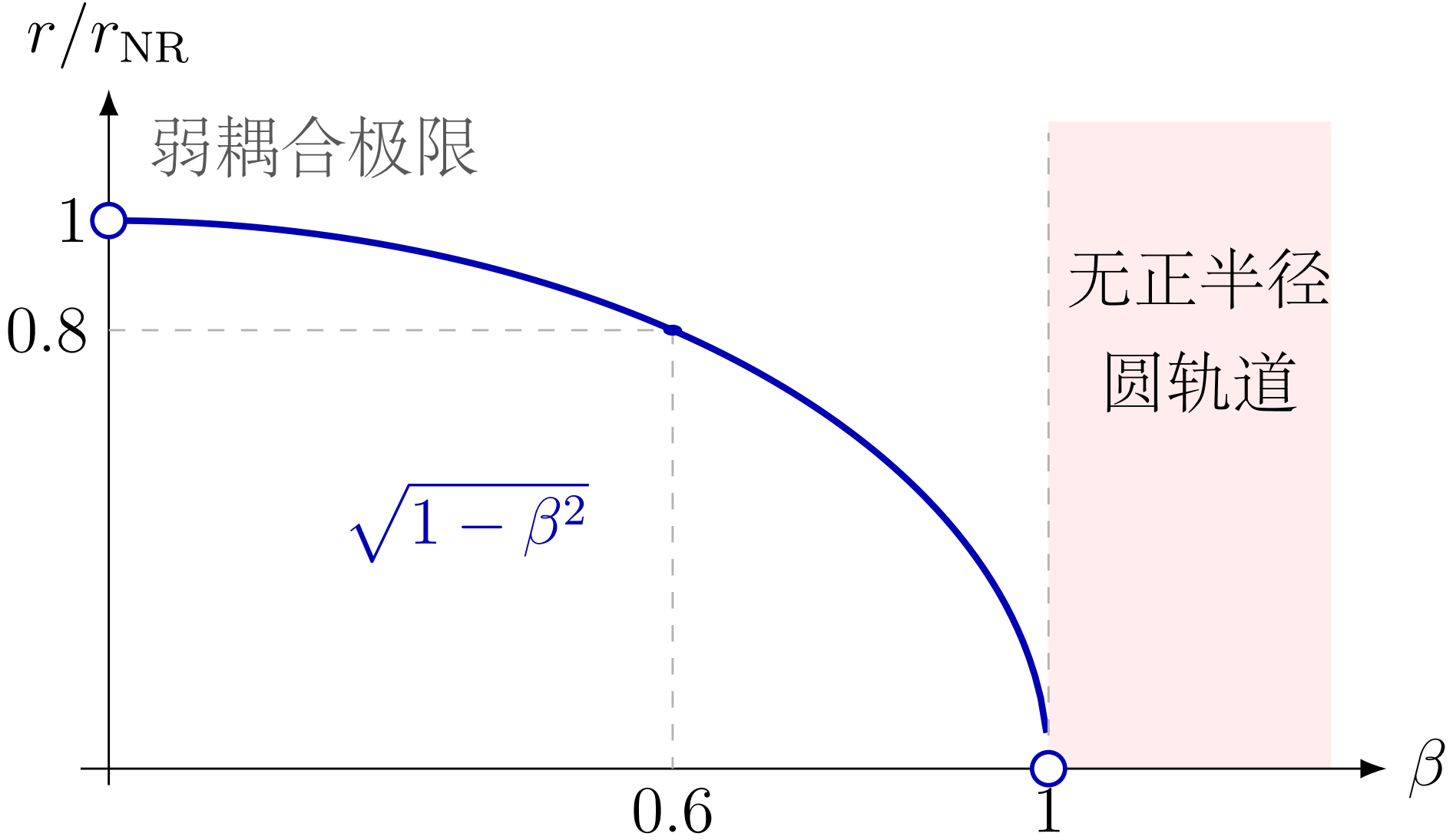

$$\beta = \alpha Z^2/(2n), \qquad r_{\mathrm{NR}} = 2n^2\hbar^2/(m_0 A)$$